^{35}P β^- decay (47.3 s) 1986Wa22

Parent: ^{35}P: E=0; J$^\pi$=1/2$^+$; T$_{1/2}$=47.3 s *8*; Q(β^-)=3988.4 *19*; %β^- decay=100.0
^{35}P-Q(β^-): From 2021Wa16.
^{35}P-J$^\pi$,T$_{1/2}$: From Adopted Levels of ^{35}P.
1986Wa22: ^{35}P was produced via the ^{36}S(t,α)^{35}P reaction. 81% enriched ^{36}S targets were bombarded with 3.4-MeV tritons from the Brookhaven National Laboratory (BNL) tandem van de Graaff facility. γ rays were detected using an intrinsic coaxial Ge detector. Measured Eγ, Iγ. Deduced levels, decay branching ratios, and log *ft*. Compared with shell model calculations.
1972Ap01: ^{35}P was produced via the ^{37}Cl(t,αp)^{35}P reaction. LiCl and NaCl targets were bombarded with 16-MeV tritons from the Los Alamos tandem Van de Graaff facility. γ rays were detected using an 85 cm^3 Ge(Li) detector. β particles were detected using an end-window gas-flow proportional counter. Measured Eγ, Iγ, and Iβ. Deduced ^{35}P T$_{1/2}$, decay branching ratios, and log *ft*. Compared with shell model calculations.
1972Go31: ^{35}P was produced via the ^{18}O(^{19}F,2p)^{35}P and ^{36}S(t,α)^{35}P reactions at the second tandem of the Brookhaven National Laboratory (BNL) tandem van de Graaff facility. ^{35}P β-delayed γ rays were detected using a 60 cm^3 Ge(Li) detector. β particles were detected in coincidence with the 1572γ using a NE102 scintillator. Measured Eγ, Iγ, and Eβ. Deduced ^{35}P T$_{1/2}$, β end-point energy, mass, decay branching ratios, and log *ft*. Compared with shell model calculations.
1971Gr53: ^{35}P was produced via the ^{37}Cl(γ,2p)^{35}P reaction. 6-g NH$_4$Cl(nat) samples were irradiated with γ rays from a bremsstrahlung spectrum at the Mainz electron linear accelerator. γ rays were detected using NaI(Tl) and Ge(Li) detectors. Measured Eγ. Deduced T$_{1/2}$ of ^{35}P and levels in ^{35}S.

^{35}S Levels

E(level)†	J$^\pi$‡	E(level)†	J$^\pi$‡	E(level)†	J$^\pi$‡
0	3/2$^+$	2717?	5/2$^+$	3597?	(1/2 to 7/2)$^+$
1572.290 *29*	1/2$^+$	2938.4 *4*	3/2$^+$	3675?	(1/2$^-$,3/2$^-$)
1991?	7/2$^-$	3421?	5/2$^+$	3802?	3/2$^-$
2348?	3/2$^-$	3558?	(3/2$^-$,5/2$^+$)		

† From 1986Wa22, unless otherwise noted.
‡ From the Adopted Levels.

β^- radiations

E(decay)	E(level)	Iβ^-†‡	Log *ft*	Comments
(186.4# *24*)	3802?	<0.03	3.5	
(313.4# *24*)	3675?	<0.02	4.4	
(391.4# *24*)	3597?	<0.02	4.7	
(430.4# *24*)	3558?	<0.11	4.2	
(567.4# *24*)	3421?	<0.02	5.5	
(1050.0 *22*)	2938.4	0.47 *3*	4.96 *4*	Iβ^-: others: <8 (1972Go31), <0.45 (1972Ap01).
(1271.4# *24*)	2717?	<0.03	7.2	Iβ^-: others: <7 (1972Go31), <0.38 (1972Ap01).
(1640.4# *24*)	2348?	<0.08	6.5	Iβ^-: others: <5 (1972Go31), <0.24 (1972Ap01).
(1997.4# *24*)	1991?	<0.08	9.1	Iβ^-: others: <2.5 (1972Go31), <0.09 (1972Ap01).
(2416.1 *22*)	1572.290	98.80 *3*	4.13 *1*	Iβ^-: other: 99 +*1*−*7* from measured γ/β ratio (1972Ap01), 100 (1972Go31).
(3988.4# *24*)	0	<0.73	>7.2	Iβ^-: other: <9 (1972Ap01).

† From 1986Wa22, unless otherwise noted.
‡ Absolute intensity per 100 decays.
# Existence of this branch is questionable.

^{35}P β^- decay (47.3 s) 1986Wa22 (continued)

$\gamma(^{35}S)$

Iγ normalization: 1.0 from Σ%I(γ to g.s.)=99.27, deduced from 100−%Iβ^-(g.s.), where %Iβ^-(g.s.)=0.73 from shell-model calculations in 1986Wa22.

E_γ[†]	I_γ[†‡]	E_i(level)	J^π_i	E_f	J^π_f	Comments
1572.252 *29*	98.80 *3*	1572.290	1/2$^+$	0	3/2$^+$	E_γ: weighted average of 1571.8 *2* (1971Gr53), 1572.2 *4* (1972Ap01), 1572.36 *24* (1972Go31), 1572.47 *25* (1972Go31), and 1572.256 *24* (1986Wa22).
2938.29 *40*	0.47 *3*	2938.4	3/2$^+$	0	3/2$^+$	

† From 1986Wa22, unless otherwise noted.
‡ Absolute intensity per 100 decays.

^{35}P β^- decay (47.3 s) 1986Wa22

Decay Scheme

Intensities: I_γ per 100 parent decays